

Louis Kang

Neural Circuits and Computations Unit

RIKEN Center for Brain Science

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<https://louiskang.group>

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POSITIONS

Unit Leader (Junior Group Leader), Neural Circuits and Computations Unit 2020–
RIKEN Center for Brain Science, Wako, Japan

Miller Postdoctoral Fellow 2017–2020
University of California, Berkeley, USA
Host departments: Physics and Helen Wills Neuroscience Institute
Host faculty: Mike DeWeese

ADJUNCT POSITIONS

Adjunct Associate Professor, Graduate School of Informatics 2021–
Kyoto University, Japan

RESEARCH STATEMENT

I study how neural computations in the hippocampal region enable us to remember events along with where they occurred. To do so, I use a variety of theoretical techniques, including neural network simulations and mathematical analysis. I aim to uncover how these computations are disrupted in neurological diseases such as Alzheimer’s disease and epilepsy, and I take inspiration from them to develop new algorithms for machine learning.

EDUCATION

MD, Perelman School of Medicine 2017
University of Pennsylvania, Philadelphia, USA

PhD, Department of Physics & Astronomy 2015
University of Pennsylvania, Philadelphia, USA
Thesis advisor: Tom Lubensky
Thesis title: *Chirality and its spontaneous symmetry breaking in two liquid crystal systems*

AB in Chemistry and Physics and Mathematics *summa cum laude* 2009
Harvard University, Cambridge, USA

PUBLICATIONS

*equal contribution †corresponding author

15. Eydam S, **Kang L**. Neural oscillations in the entorhinal grid network from function to pathology. In *Frontiers of Neural Oscillations Research*, edited by Ohki T. *Neuromethods*, New York: Humana (in press).

14. Wang S, **Kang L**, Salamon P, Wang X, Uchida N, Araoka F, Aida T, Dogic Z, Ishida Y[†]. Stimuli-responsive self-regulating assembly of chiral colloids for robust size and shape control. *Nat Commun* 15, 9891 (2024). doi:10.1038/s41467-024-54217-x.
13. Eydam S[†], Franović I, **Kang L**. Control of seizure-like dynamics in neuronal populations with excitability adaptation related to ketogenic diet. *Chaos* 34, 053128 (2024). doi:10.1063/5.0180954.
12. **Kang L**[†], Toyozumi T. Distinguishing examples while building concepts in hippocampal and artificial networks. *Nat Commun* 15, 647 (2024). doi:10.1038/s41467-024-44877-0.
11. **Kang L**[†], Toyozumi T. A Hopfield-like model with complementary encodings of memories. *Phys Rev E* 108, 054410 (2023). doi:10.1103/PhysRevE.108.054410.
10. Wang R, **Kang L**[†]. Multiple bumps can enhance robustness to noise in continuous attractor networks. *PLOS Comput Biol* 18, e1010547 (2022). doi:10.1371/journal.pcbi.1010547.
9. **Kang L**[†], Xu B, Morozov D. Evaluating state space discovery by persistent cohomology in the spatial representation system. *Front Comput Neurosci* 15, 616748 (2021). doi:10.3389/fncom.2021.616748.
8. **Kang L**[†], DeWeese MR. Replay as wavefronts and theta sequences as bump oscillations in a grid cell attractor network. *eLife* 8, e46351 (2019). doi:10.7554/eLife.46351.
7. **Kang L**[†], Balasubramanian V. A geometric attractor mechanism for self-organization of entorhinal grid modules. *eLife* 8, e46687 (2019). doi:10.7554/eLife.46687.
6. **Kang L**[†], Lubensky TC. Chiral twist drives raft formation and organization in membranes composed of rod-like particles. *Proc Natl Acad Sci USA* 114, E19 (2017). doi:10.1073/pnas.1613732114.
5. **Kang L**[†], Gibaud T, Dogic Z, Lubensky TC. Entropic forces stabilize diverse emergent structures in colloidal membranes. *Soft Matter* 12, 386 (2016). doi:10.1039/C5SM02038G.
4. Davidson ZS*, **Kang L***, Jeong J*[†], Still T, Collings PJ, Lubensky TC, Yodh AG. Chiral structures and defects of lyotropic chromonic liquid crystals induced by saddle-splay elasticity. *Phys Rev E* 91, 050501 (2015). doi:10.1103/PhysRevE.91.050501.
3. Jeong J*[†], **Kang L***, Davidson ZS, Collings PJ, Lubensky TC, Yodh AG. Chiral structures from achiral liquid crystals in cylindrical capillaries. *Proc Natl Acad Sci USA* 112, E1837 (2015). doi:10.1073/pnas.1423220112.
2. Idema T, Dubuis JO, **Kang L**, Manning ML, Nelson PC, Lubensky TC, Liu AJ[†]. The syncytial *Drosophila* embryo as a mechanically excitable medium. *PLOS ONE* 8, e77216 (2013). doi:10.1371/journal.pone.0077216.
1. Heo M, **Kang L**, Shakhnovich EI[†]. Emergence of species in evolutionary “simulated annealing”. *Proc Natl Acad Sci USA* 106, 1869 (2009). doi:10.1073/pnas.0809852106.

GRANTS, AWARDS, AND HONORS

KAKENHI Grant-in-Aid for Transformative Research Areas (A)

2026–2030

Japan Society for the Promotion of Science

Multi-project group title: *Dream Science (26A303)*

Project role: PI

Project title: *Computational principles of dreaming in brains and machines (A03)*

KAKENHI Grant-in-Aid for Early-Career Scientists 2022–2025
 Japan Society for the Promotion of Science
 Project role: PI
 Project title: *The influence of attractor topology on seizure initiation in the hippocampal region (22K15209)*

Collaborative Research Travel Grant 2019–2020
 Burroughs Wellcome Fund
 Project role: PI
 Project title: *Complementary input pathways enhance associative memory in a model of CA3*

Miller Research Fellowship 2017–2020
 University of California, Berkeley

Mary Ellis Bell Prize 2016
 University of Pennsylvania, Perelman School of Medicine
 “This prize is given to a student in the School of Medicine who is engaged in noteworthy research in any field related to medicine.”

Werner Teutsch Memorial Prize 2012
 University of Pennsylvania, Department of Physics & Astronomy
 “Awarded annually to the graduate student who, by his or her performance in the first year courses, shows the most promise for outstanding achievement in research.”

Medical Scientist Training Program 2009–2017
 National Institutes of Health (USA), awarded through the University of Pennsylvania

Phi Beta Kappa 2009
 Harvard University

INVITED TALKS *competitively selected —————

Sakana AI, Tokyo, Japan 2025

University of California, San Francisco, USA 2025
 Neuroscience Graduate Program Retreat

World Sleep Congress, Singapore 2025

Korean Society for Brain and Neural Sciences Annual Meeting, Gyeongju, Korea 2024

Simons Institute for the Theory of Computing, Berkeley, USA 2024
 Summer Cluster in AI, Psychology, and Neuroscience

Simons Collaboration for the Global Brain, online 2020
 West Coast Postdoc/Student Meeting

Bernstein Conference*, Berlin, Germany 2018

Computational Neuroscience Meeting*, Seattle, USA 2018

SCIENTIFIC COMMUNITY INVOLVEMENT —————

Conference Co-organizer, RIKEN CBS Co-Creation International Conference 2025
Intelligent neural systems: what, how, and why?

Speakers: Shun-ichi Amari, Anne Collins, Tomoki Fukai, Pulin Gong, Ryo Karakida, Taegon Kim, Máté Lengyel, Anna Levina, Yukie Nagai, Mizuki Oka, Masafumi Oizumi, Tomoyo Shiramatsu

Workshop Co-organizer, Computational and Systems Neuroscience (Cosyne) 2023
Seeking universality while celebrating heterogeneity among biological attractor networks

Speakers: Nicolas Brunel, Yoram Burak, Joanna Chang, Albert Compte, Kayvon Daie, Laura Driscoll, Tatiana Engel, Ila Fiete, Kevin Franks, Lisa Giocomo, Yi Gu, Christiane Linster, Luca Mazzucato, Adit Radhakrishnan, Valentin Schmutz, Misha Tsodyks, Dan Turner-Evans, Kechen Zhang

Grant Reviewer

Swiss National Science Foundation

Journal Reviewer

Nature Communications, PLOS Computational Biology, Journal of Neuroscience, Neural Computation, Physical Review E, Frontiers in Computational Neuroscience, Neural Networks

Conference Reviewer

Computational and Systems Neuroscience (Cosyne)

Prize Juror

Hidden Beauty of Science Contest organized by Ambassade de France au Japon

TEACHING

An introduction to computational neuroscience 2022–
 RIKEN Center for Brain Science, Brain Science Training Program
 Two-hour lecture for graduate students once a year

Recurrent neural networks in the brain 2021–
 Kyoto University, Graduate School of Informatics
 Three-hour lecture for graduate students once a year

REFERENCES

Taro Toyozumi

Theoretical collaborator

RIKEN Center for Brain Science

Neural Adaptation and Computation Lab

taro.toyoizumi@riken.jp

Yoram Burak

External expert in attractor networks

Hebrew University of Jerusalem

Safra Center for Brain Sciences

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Tom Lubensky

PhD advisor

University of Pennsylvania

Department of Physics & Astronomy

tom@physics.upenn.edu

Tom McHugh

Experimental collaborator

RIKEN Center for Brain Science

Circuit and Behavioral Physiology Lab

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Mike DeWeese

Postdoctoral advisor

University of California, Berkeley

Redwood Center for Theoretical Neuroscience

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Vijay Balasubramanian

Research mentor

University of Pennsylvania

Department of Physics & Astronomy

vijay@physics.upenn.edu